

Solução do Problema 80bau3b - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: 80bau3b
- Número de Variáveis: 9799
- Número de Restrições: 2262
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 1.377e-14
- Valor Primal: 9.872e+05
- Valor Dual: 1.130e+06
- Gap: 2.986e-22
- Número de Iterações: 43

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema 80bau3b

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	5.311e+02	50	1.327e+00
2	2.868e+02	56	1.327e+00
3	5.447e+02	62	1.327e+00
4	8.955e+02	73	1.327e+00
5	1.811e+03	94	1.327e+00
6	1.642e+03	102	5.952e-01
7	1.076e+03	134	3.848e+00
8	3.587e+02	135	1.534e+00
9	3.459e+02	136	1.381e+00
10	7.684e+02	137	1.534e+00
11	4.000e-04	149	1.534e+00
12	4.000e-04	150	1.381e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	4.000e-04	151	1.534e+00
14	4.000e-04	167	1.104e+00
15	4.000e-04	168	1.087e+00
16	4.000e-04	169	1.087e+00
17	4.000e-04	189	7.352e-01
18	4.000e-04	190	1.010e+00
19	4.000e-04	192	1.534e+00
20	4.000e-04	193	1.381e+00
21	4.000e-04	194	1.534e+00
22	4.000e-04	200	1.575e-01
23	4.000e-04	214	4.808e+00
24	4.000e-04	219	1.343e-01
25	4.000e-04	220	3.561e+00
26	4.000e-04	221	1.534e+00
27	4.000e-04	222	1.381e+00
28	4.000e-04	223	1.534e+00
29	4.000e-04	228	1.055e+00
30	4.000e-04	229	1.669e+00
31	4.000e-04	243	4.771e+00
32	4.000e-04	248	1.534e+00
33	4.000e-04	249	1.381e+00
34	4.000e-04	250	1.534e+00
35	4.000e-04	257	5.047e-01
36	4.000e-04	270	4.704e+00
37	4.000e-04	276	5.511e+00
38	4.000e-04	277	5.810e+00
39	4.000e-04	278	1.534e+00
40	4.000e-04	279	1.381e+00
41	4.000e-04	280	1.534e+00
42	4.000e-04	283	7.168e+00
43	4.000e-04	284	4.931e+00
44	4.000e-04	289	1.722e+00
45	4.000e-04	290	4.831e+00
46	7.000e-05	291	4.036e+00
47	3.000e-02	292	5.937e+00
48	9.000e-02	293	5.124e+00
49	7.000e-05	294	7.112e+00
51	6.482e-02	306	7.416e-01
52	3.700e-01	307	1.317e+00
53	4.000e-04	308	6.732e-01
54	4.000e-04	309	1.854e+00
55	7.000e-05	310	1.142e+00
57	1.100e-01	311	2.979e+00
58	3.700e-01	315	4.911e-01
59	4.000e-04	318	1.695e-01
60	4.000e-04	319	1.307e+00
61	7.000e-05	324	6.769e+00
63	1.100e-01	329	1.534e+00
64	3.700e-01	330	1.381e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
65	4.000e-04	331	1.534e+00
66	4.000e-04	349	2.825e+00
67	3.697e+00	358	2.280e-03
68	5.000e-02	359	1.150e-03
69	1.800e-01	361	1.130e-03
70	3.000e+01	364	9.034e+00
71	4.000e-04	367	1.130e-03
72	3.635e+00	372	1.542e+00
74	8.000e-02	378	8.774e-01
75	2.700e-01	381	2.680e+00
76	4.000e-04	382	1.939e+00
77	4.000e-04	383	9.890e-01
78	7.000e-05	391	2.280e-03
79	1.000e-02	392	1.175e+00
80	4.000e-02	393	3.438e+00
81	4.000e-04	394	4.678e+00
82	4.000e-04	395	3.116e+00
83	3.697e+00	396	2.883e+00
84	1.000e-02	397	2.669e+00
85	4.000e-02	401	1.140e-03
86	4.000e-04	402	1.054e+00
87	4.000e-04	403	1.866e+00
88	3.697e+00	404	7.400e-01
89	1.000e-02	407	2.466e+00
90	2.000e-02	413	2.280e-03
91	4.000e-04	415	1.150e-03
92	4.000e-04	416	1.696e+00
93	7.000e-05	417	1.130e-03
95	3.000e-02	420	2.280e-03
96	9.000e-02	421	1.759e-02
97	1.551e+03	422	6.331e-01
98	3.660e+02	426	2.228e+00
99	2.699e+03	430	2.280e-03
100	3.690e+02	431	1.150e-03
101	7.925e+03	432	7.877e-01
103	5.860e+02	435	2.172e+00
105	2.943e+03	436	2.369e+00
107	2.720e+02	437	1.130e-03
108	3.770e+02	441	5.004e+00
109	1.232e+03	442	2.280e-03
110	2.050e+02	444	2.887e+00
111	2.500e+02	445	4.986e-01
113	4.329e+00	449	8.004e-03
114	1.656e+00	451	2.206e+01
115	8.712e+00	452	1.136e+01
116	4.449e+01	453	4.232e+00
117	8.388e+01	454	1.736e+00
120	2.760e+00	455	1.023e+00
121	4.000e-04	458	4.034e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
122	4.000e-04	459	2.899e+00
123	4.000e-04	460	1.003e+01
124	8.522e+01	462	2.463e+01
125	1.272e+01	463	1.232e+01
127	4.000e-04	464	4.105e+00
128	4.000e-04	465	1.232e+00
129	3.007e-03	466	4.104e-01
130	4.800e+00	468	4.107e-01
131	3.900e+00	469	1.232e+00
133	3.602e+00	470	4.106e+00
134	8.160e+00	471	1.232e+01
135	3.552e+00	472	2.463e+01
138	2.280e+00	473	1.232e+01
139	1.363e+01	474	4.105e+00
142	5.654e+01	475	1.232e+00
143	2.880e+01	476	4.104e-01
145	4.000e-04	478	4.107e-01
146	4.000e-04	479	1.232e+00
147	3.120e+00	480	4.106e+00
149	1.440e+00	481	1.232e+01
152	3.552e+00	482	2.544e+01
153	4.440e+01	483	1.272e+01
156	4.000e-04	484	4.239e+00
157	4.000e-04	485	1.272e+00
158	4.000e-04	486	4.238e-01
159	9.842e+01	488	4.241e-01
161	6.000e+01	489	1.272e+00
162	4.000e-04	490	4.240e+00
163	3.007e-03	491	1.272e+01
164	2.040e+01	492	2.565e+01
165	7.200e+01	493	1.282e+01
166	3.482e+01	494	4.274e+00
168	6.000e-01	495	1.282e+00
170	7.227e-01	496	4.273e-01
171	5.256e+00	498	4.276e-01
172	9.600e+00	499	1.282e+00
173	3.972e+01	500	4.275e+00
175	3.120e+01	501	1.282e+01
177	4.000e-04	502	1.268e+01
178	4.000e-04	503	6.339e+00
179	4.000e-04	504	2.111e+00
180	1.175e+02	505	6.310e-01
182	4.000e-04	506	2.082e-01
183	4.000e-04	508	2.146e-01
184	3.007e-03	509	6.374e-01
185	1.440e+01	510	2.117e+00
186	1.178e+02	511	6.345e+00
187	2.340e+00	512	7.539e+00
188	4.795e+00	513	3.768e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
189	6.648e+01	514	1.254e+00
191	1.000e-04	515	3.739e-01
192	1.999e+01	516	1.226e-01
195	4.795e+00	518	1.288e-01
196	2.438e+01	519	3.802e-01
197	3.048e+01	520	1.260e+00
198	1.008e+01	521	3.774e+00
199	1.440e+01	522	1.277e+01
202	6.475e+01	523	6.384e+00
203	1.728e+01	524	2.126e+00
204	2.977e+02	525	6.355e-01
205	4.000e-04	526	2.097e-01
206	4.000e-04	528	2.161e-01
207	4.000e-04	529	6.419e-01
208	4.426e+02	530	2.132e+00
209	2.640e+01	531	6.390e+00
211	4.000e-04	533	3.089e+01
212	4.000e-04	534	1.545e+01
213	3.007e-03	535	5.147e+00
214	4.111e+01	536	1.542e+00
215	3.802e+01	537	5.122e-01
216	4.795e+00	539	5.178e-01
217	6.643e+01	540	1.548e+00
218	1.200e+00	541	5.152e+00
221	6.504e+00	542	1.545e+01
224	3.840e+00	543	2.553e+01
225	1.898e+01	544	1.277e+01
226	9.840e-01	545	4.256e+00
227	2.102e+02	546	1.277e+00
228	1.200e+01	547	4.255e-01
230	2.705e+01	549	4.256e-01
231	8.256e+00	550	1.277e+00
233	4.000e-04	551	4.256e+00
234	4.000e-04	552	1.277e+01
235	4.000e-04	555	1.384e+01
236	3.428e+02	556	6.920e+00
237	2.712e+01	557	2.307e+00
238	2.304e+01	558	6.920e-01
240	2.000e+01	559	2.307e-01
241	4.000e-04	561	2.307e-01
242	7.667e+01	562	6.920e-01
243	3.223e+01	563	2.307e+00
244	4.000e-04	564	6.920e+00
245	4.795e+00	566	1.445e+01
247	2.397e+00	567	7.223e+00
248	1.752e+00	568	2.408e+00
251	5.028e+01	569	7.223e-01
252	2.132e+02	570	2.408e-01
253	1.464e+00	572	2.408e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
254	7.680e-01	573	7.223e-01
255	6.156e+01	574	2.408e+00
256	2.256e+01	575	7.223e+00
258	1.320e+01	579	3.304e+00
260	4.000e-04	580	1.661e+00
261	4.000e-04	581	5.659e-01
262	4.000e-04	582	1.826e-01
263	1.843e+02	583	7.306e-02
264	2.351e+02	585	3.645e-02
265	4.224e+01	586	1.460e-01
267	4.000e-04	587	5.292e-01
268	4.000e-04	588	1.624e+00
269	3.007e-03	589	3.472e+00
270	1.152e+01	590	1.733e+00
271	7.200e+00	591	5.746e-01
272	4.000e-04	592	1.690e-01
273	3.900e+00	593	5.316e-02
274	4.795e+00	595	6.272e-02
275	1.000e-04	596	1.786e-01
276	2.436e+01	597	5.842e-01
278	1.378e+01	598	1.743e+00
281	7.440e-01	599	2.866e+00
284	3.038e+01	600	1.442e+00
286	4.000e-04	601	4.929e-01
287	4.000e-04	602	1.607e-01
288	4.000e-04	603	6.577e-02
289	5.448e+01	605	2.915e-02
290	8.090e+01	606	1.241e-01
292	4.080e+00	607	4.562e-01
295	4.000e-04	608	1.405e+00
296	4.000e-04	610	3.586e+00
297	3.007e-03	611	1.802e+00
298	2.976e+00	612	6.129e-01
299	4.656e+00	613	1.967e-01
300	1.062e+01	614	7.776e-02
301	4.334e+01	616	4.115e-02
302	9.984e+00	617	1.601e-01
303	9.979e+01	618	5.762e-01
304	1.620e+01	619	1.765e+00
306	3.600e+01	622	2.220e+01
308	4.968e+00	623	1.110e+01
309	9.600e+00	624	3.697e+00
312	4.000e-04	625	1.107e+00
313	4.000e-04	626	3.668e-01
314	4.000e-04	628	3.733e-01
315	7.920e+00	629	1.113e+00
316	1.128e+01	630	3.704e+00
317	9.120e+00	631	1.110e+01
318	3.043e+01	632	2.269e+01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
320	6.000e+01	633	1.134e+01
321	4.000e-04	634	3.779e+00
322	3.007e-03	635	1.131e+00
323	2.400e+01	636	3.750e-01
325	4.000e-04	638	3.815e-01
326	7.181e+00	639	1.138e+00
327	6.528e+00	640	3.786e+00
328	8.976e+00	641	1.135e+01
329	8.184e+00	642	2.178e+01
332	2.369e+01	643	1.089e+01
333	1.680e+01	644	3.628e+00
334	4.184e+01	645	1.086e+00
336	6.000e+00	646	3.599e-01
337	2.232e+01	648	3.664e-01
338	6.000e+00	649	1.093e+00
339	1.142e+02	650	3.635e+00
340	4.000e-04	651	1.090e+01
341	4.000e-04	653	8.004e-03
342	4.000e-04	655	2.371e+01
343	1.968e+02	656	1.185e+01
344	7.227e-01	657	3.952e+00
346	6.000e+01	658	1.185e+00
347	4.000e-04	659	3.951e-01
348	3.007e-03	661	3.952e-01
349	3.384e+00	662	1.186e+00
350	3.281e+01	663	3.952e+00
351	2.100e+01	664	1.185e+01
352	2.340e+00	666	2.611e+01
353	4.795e+00	667	1.305e+01
354	7.227e-01	668	4.350e+00
355	7.681e+01	669	1.303e+00
357	1.656e+00	670	4.328e-01
362	2.112e+00	672	4.378e-01
363	4.080e+00	673	1.308e+00
365	3.854e+01	674	4.355e+00
366	1.594e+01	675	1.306e+01
368	1.709e+01	676	2.611e+01
369	1.061e+01	677	1.305e+01
370	3.912e+00	678	4.350e+00
373	2.902e+01	679	1.303e+00
374	5.479e+01	680	4.328e-01
375	1.122e+02	682	4.378e-01
376	9.912e+00	683	1.308e+00
377	3.742e+01	684	4.355e+00
379	3.672e+00	685	1.306e+01
384	4.234e+01	686	2.702e+01
385	4.824e+00	687	1.351e+01
386	1.658e+01	688	4.501e+00
387	3.504e+00	689	1.348e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
388	2.928e+01	690	4.478e-01
389	2.398e+01	692	4.529e-01
398	3.626e+01	693	1.354e+00
399	1.918e+01	694	4.506e+00
400	4.560e+00	695	1.351e+01
408	1.337e+01	696	2.726e+01
409	6.701e+01	697	1.363e+01
410	1.183e+01	698	4.542e+00
411	3.495e+01	699	1.361e+00
414	1.807e+01	700	4.519e-01
423	2.609e+01	702	4.570e-01
427	3.456e+00	703	1.366e+00
439	2.038e+01	704	4.547e+00
440	1.550e+01	705	1.364e+01
446	1.058e+02	706	3.349e+01
447	3.384e+00	707	1.674e+01
448	2.511e+01	708	5.579e+00
450	1.840e-01	709	1.671e+00
456	9.499e+00	710	5.546e-01
457	1.000e-03	712	5.619e-01
461	1.000e-03	713	1.678e+00
467	4.140e+00	714	5.586e+00
477	3.403e+01	715	1.675e+01
487	1.582e+01	716	2.725e+01
497	1.151e+02	717	1.362e+01
507	8.517e+01	718	4.538e+00
517	2.378e+02	719	1.359e+00
527	1.530e+02	720	4.505e-01
532	4.015e-01	722	4.579e-01
538	1.968e+02	723	1.366e+00
548	6.764e+01	724	4.545e+00
553	4.644e+00	725	1.363e+01
554	2.934e-01	726	3.368e+01
560	1.767e+00	727	1.684e+01
565	3.227e+01	728	5.610e+00
571	4.304e+00	729	1.680e+00
576	5.792e-01	730	5.577e-01
577	1.000e-03	732	5.651e-01
578	4.430e+00	733	1.688e+00
584	1.671e+02	734	5.617e+00
594	1.280e+02	735	1.684e+01
604	2.272e+02	737	3.163e+01
609	9.604e+01	738	1.582e+01
615	2.088e+02	739	5.270e+00
620	9.063e+00	740	1.580e+00
621	1.355e+01	741	5.251e-01
627	2.955e+00	743	5.294e-01
637	8.036e+00	744	1.584e+00
647	3.475e+01	745	5.275e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
652	1.417e+01	746	1.582e+01
654	7.910e-03	747	2.631e+01
660	8.652e-01	748	1.315e+01
665	1.000e-03	749	4.383e+00
671	1.894e+01	750	1.313e+00
681	1.776e+01	751	4.362e-01
691	1.247e+02	753	4.410e-01
701	4.926e+01	754	1.318e+00
711	9.484e+01	755	4.388e+00
721	6.871e+01	756	1.316e+01
731	9.462e+01	759	1.509e+01
736	2.409e-01	760	7.547e+00
742	2.350e+02	761	2.516e+00
752	3.758e+01	762	7.546e-01
757	1.982e+00	763	2.515e-01
758	7.794e+00	765	2.516e-01
764	3.801e+00	766	7.547e-01
769	4.663e+01	767	2.516e+00
775	9.770e+00	768	7.547e+00
780	1.981e-01	770	1.565e+01
781	1.000e-03	771	7.826e+00
782	5.406e+00	772	2.609e+00
788	3.886e+02	773	7.826e-01
798	2.396e+01	774	2.608e-01
808	1.023e+02	776	2.609e-01
813	4.133e+00	777	7.826e-01
819	3.422e+02	778	2.609e+00
824	2.968e+01	779	7.826e+00
825	1.290e+01	783	3.768e+00
831	1.900e+01	784	1.884e+00
841	6.797e+01	785	6.288e-01
851	8.061e+00	786	1.894e-01
856	3.998e+01	787	6.383e-02
858	2.021e-01	789	6.172e-02
864	1.184e+01	790	1.873e-01
869	1.000e-03	791	6.267e-01
875	4.073e+01	792	1.882e+00
885	6.215e+01	793	3.629e+00
895	1.201e+02	794	1.812e+00
905	8.989e+01	795	6.013e-01
915	1.989e+02	796	1.774e-01
925	1.392e+02	797	5.631e-02
935	1.538e+02	799	6.481e-02
940	1.124e+00	800	1.859e-01
946	3.832e+02	801	6.098e-01
956	1.061e+02	802	1.821e+00
961	2.283e+00	803	3.438e+00
962	1.384e+01	804	1.719e+00
968	3.775e+00	805	5.738e-01

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
973	1.230e+02	806	1.729e-01
979	1.039e+01	807	5.833e-02
984	5.949e-01	809	5.622e-02
985	1.169e+01	810	1.708e-01
986	1.376e+01	811	5.717e-01
992	7.742e+02	812	1.717e+00
1002	4.356e+01	814	4.052e+00
1012	5.497e+02	815	2.025e+00
1017	4.866e+01	816	6.741e-01
1023	9.998e+02	817	2.013e-01
1028	3.883e+02	818	6.617e-02
1029	3.899e+01	820	6.894e-02
1035	4.088e+01	821	2.040e-01
1045	1.098e+02	822	6.769e-01
1055	4.768e+01	823	2.028e+00
1060	5.177e+01	826	2.345e+01
1062	1.587e+00	827	1.172e+01
1068	7.179e+01	828	3.906e+00
1073	1.000e-03	829	1.170e+00
1079	1.391e+01	830	3.883e-01
1089	5.155e+01	832	3.935e-01
1099	1.088e+02	833	1.175e+00
1109	1.912e+02	834	3.911e+00
1119	2.128e+02	835	1.173e+01
1129	3.442e+02	836	2.364e+01
1139	2.580e+02	837	1.182e+01
1144	8.833e-01	838	3.938e+00
1150	4.852e+02	839	1.180e+00
1160	7.183e+01	840	3.915e-01
1165	2.156e+00	842	3.967e-01
1166	8.026e+00	843	1.185e+00
1172	1.058e+01	844	3.943e+00
1177	9.758e+01	845	1.182e+01
1183	1.243e+01	846	2.292e+01
1188	7.951e-01	847	1.146e+01
1189	9.370e+01	848	3.818e+00
1190	1.067e+01	849	1.144e+00
1196	6.273e+02	850	3.795e-01
1206	1.246e+02	852	3.847e-01
1216	7.737e+02	853	1.149e+00
1221	1.441e+02	854	3.823e+00
1227	1.045e+03	855	1.146e+01
1232	1.651e+02	857	7.996e-03
1233	4.317e+01	859	2.404e+01
1239	1.360e+01	860	1.202e+01
1249	4.605e+01	861	4.007e+00
1259	3.357e+01	862	1.202e+00
1264	1.187e+02	863	4.006e-01
1266	3.268e+00	865	4.007e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1272	6.558e+01	866	1.202e+00
1277	1.000e-03	867	4.006e+00
1283	1.352e+01	868	1.202e+01
1293	9.859e+01	870	2.617e+01
1303	4.355e+01	871	1.308e+01
1313	4.352e+02	872	4.359e+00
1323	3.753e+02	873	1.306e+00
1333	6.892e+02	874	4.336e-01
1343	5.659e+02	876	4.387e-01
1348	1.124e+00	877	1.311e+00
1354	9.629e+02	878	4.364e+00
1364	1.943e+02	879	1.309e+01
1369	1.068e+01	880	2.617e+01
1370	1.053e+02	881	1.308e+01
1376	3.075e+01	882	4.359e+00
1381	1.914e+02	883	1.306e+00
1387	5.389e+01	884	4.336e-01
1392	1.933e+00	886	4.387e-01
1393	2.465e+01	887	1.311e+00
1394	2.276e+01	888	4.364e+00
1400	8.320e+02	889	1.309e+01
1410	3.754e+02	890	2.711e+01
1420	1.838e+03	891	1.356e+01
1425	3.746e+02	892	4.517e+00
1431	9.234e+02	893	1.353e+00
1436	1.460e+02	894	4.494e-01
1437	6.419e+01	896	4.545e-01
1443	1.115e+01	897	1.358e+00
1453	1.284e+02	898	4.522e+00
1463	5.350e+01	899	1.356e+01
1468	1.418e+02	900	2.735e+01
1470	1.144e+00	901	1.368e+01
1476	1.330e+02	902	4.557e+00
1481	1.000e-03	903	1.365e+00
1487	1.602e+01	904	4.535e-01
1497	1.126e+02	906	4.584e-01
1507	1.094e+02	907	1.370e+00
1517	3.981e+02	908	4.562e+00
1527	3.320e+02	909	1.368e+01
1537	7.305e+02	910	3.258e+01
1547	4.300e+02	911	1.631e+01
1552	1.606e+00	912	5.463e+00
1558	9.891e+02	913	1.667e+00
1568	1.304e+02	914	5.817e-01
1573	8.716e+00	916	5.032e-01
1574	7.359e+01	917	1.588e+00
1580	8.016e+01	918	5.385e+00

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema 80bau3b

Coordenada y	Valor y (Dual)
12	1.534e+00
42	1.534e+00
46	3.587e+01
47	3.499e+01
48	3.499e+01
49	3.499e+01
50	4.105e+01
51	4.105e+01
52	4.239e+01
53	4.274e+01
54	2.114e+01
55	1.257e+01
56	2.129e+01
57	1.690e+01
58	5.150e+01
59	4.256e+01
60	2.307e+01
61	3.695e+01
62	2.307e+01
63	3.695e+01
64	2.408e+01
65	2.139e+01
66	8.689e+01
67	3.695e+01
68	5.457e+00
69	5.799e+00
70	4.727e+00
71	4.727e+00
72	5.927e+00
73	3.587e+01
74	3.701e+01
75	3.783e+01
76	3.632e+01
77	3.790e+01
78	3.952e+01
79	3.952e+01
80	3.952e+01
81	4.353e+01
82	4.353e+01
83	4.504e+01
84	4.545e+01
85	5.583e+01
Continua na próxima página	

Coordenada y	Valor y (Dual)
86	4.542e+01
87	5.614e+01
88	5.313e+01
89	5.273e+01
90	4.386e+01
91	2.516e+01
92	3.877e+01
93	2.516e+01
94	3.877e+01
95	2.609e+01
96	2.360e+01
97	5.697e+01
98	3.877e+01
99	6.277e+00
100	6.060e+00
101	5.727e+00
102	5.727e+00
103	6.757e+00
104	3.790e+01
105	3.909e+01
106	3.941e+01
107	3.821e+01
108	3.782e+01
109	4.007e+01
110	4.007e+01
111	4.007e+01
112	4.362e+01
113	4.362e+01
114	4.520e+01
115	4.560e+01
116	5.420e+01
117	3.568e+01
118	5.760e+01
119	4.956e+01
120	5.349e+01
121	4.439e+01
122	2.541e+01
123	3.869e+01
124	2.541e+01
125	3.869e+01
126	2.649e+01
127	2.352e+01
128	4.858e+01
129	3.869e+01
130	6.261e+00
131	6.224e+00
132	5.661e+00
133	5.661e+00
134	6.671e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
135	3.782e+01
136	3.972e+01
137	4.041e+01
138	3.853e+01
139	3.779e+01
140	3.321e+01
141	3.321e+01
142	3.321e+01
143	4.316e+01
144	4.316e+01
145	4.456e+01
146	4.490e+01
147	4.042e+01
148	2.954e+01
149	4.143e+01
150	3.624e+01
151	5.265e+01
152	4.583e+01
153	2.625e+01
154	3.869e+01
155	2.625e+01
156	3.869e+01
157	2.777e+01
158	2.352e+01
159	4.233e+01
160	3.869e+01
161	5.400e+00
162	6.380e+00
163	4.900e+00
164	4.900e+00
165	5.620e+00
166	3.779e+01
167	4.192e+01
168	4.136e+01
169	3.884e+01
170	3.772e+01
171	5.201e+01
172	5.201e+01
173	5.201e+01
174	4.361e+01
175	4.361e+01
176	4.518e+01
177	4.559e+01
178	3.861e+01
179	2.990e+01
180	3.782e+01
181	3.470e+01
182	5.237e+01
183	4.615e+01
Continua na próxima página	

Coordenada y	Valor y (Dual)
184	2.671e+01
185	3.862e+01
186	2.671e+01
187	3.862e+01
188	2.815e+01
189	2.390e+01
190	5.231e+01
191	3.862e+01
192	4.829e+00
193	6.007e+00
194	4.529e+00
195	4.529e+00
196	5.069e+00
197	3.772e+01
198	3.922e+01
199	3.909e+01
200	3.815e+01
201	3.737e+01
202	3.620e+01
203	3.620e+01
204	3.620e+01
205	4.123e+01
206	4.123e+01
207	4.211e+01
208	4.234e+01
209	4.091e+01
210	3.009e+01
211	4.187e+01
212	3.593e+01
213	5.213e+01
214	4.289e+01
215	2.539e+01
216	3.828e+01
217	2.539e+01
218	3.828e+01
219	2.604e+01
220	2.407e+01
221	4.953e+01
222	3.828e+01
223	5.097e+00
224	6.039e+00
225	4.817e+00
226	4.817e+00
227	5.247e+00
228	3.737e+01
229	3.948e+01
230	3.929e+01
231	3.785e+01
232	3.750e+01
Continua na próxima página	

Coordenada y	Valor y (Dual)
233	4.681e+01
234	4.681e+01
235	4.681e+01
236	4.096e+01
237	4.096e+01
238	4.171e+01
239	4.189e+01
240	4.943e+01
241	4.130e+01
242	5.168e+01
243	4.651e+01
244	5.158e+01
245	4.409e+01
246	2.545e+01
247	3.841e+01
248	2.545e+01
249	3.841e+01
250	2.634e+01
251	2.369e+01
252	5.859e+01
253	3.841e+01
254	5.094e+00
255	4.364e+00
256	4.364e+00
257	4.364e+00
258	5.134e+00
259	3.750e+01
260	3.942e+01
261	3.942e+01
262	3.804e+01
263	3.730e+01
264	3.007e+01
265	3.007e+01
266	3.007e+01
267	4.086e+01
268	4.086e+01
269	4.162e+01
270	4.185e+01
271	3.784e+01
272	2.967e+01
273	3.995e+01
274	3.606e+01
275	5.180e+01
276	4.248e+01
277	2.493e+01
278	3.820e+01
279	2.493e+01
280	3.820e+01
281	2.557e+01
Continua na próxima página	

Coordenada y	Valor y (Dual)
282	2.377e+01
283	4.936e+01
284	3.820e+01
285	4.806e+00
286	6.081e+00
287	4.386e+00
288	4.386e+00
289	4.986e+00
290	3.730e+01
291	3.973e+01
292	3.954e+01
293	3.797e+01
294	3.690e+01
295	2.273e+01
296	2.273e+01
297	2.273e+01
298	4.206e+01
299	4.206e+01
300	4.340e+01
301	4.375e+01
302	2.407e+01
303	1.359e+01
304	2.656e+01
305	1.918e+01
306	5.142e+01
307	4.313e+01
308	2.566e+01
309	3.787e+01
310	2.566e+01
311	3.787e+01
312	2.655e+01
313	2.414e+01
314	6.241e+01
315	3.787e+01
316	4.629e+00
317	4.255e+00
318	4.249e+00
319	4.249e+00
320	4.729e+00
321	3.690e+01
322	3.842e+01
323	3.810e+01
324	3.716e+01
325	3.587e+01
326	4.953e+01
327	4.953e+01
328	4.953e+01
329	4.105e+01
330	4.105e+01
Continua na próxima página	

Coordenada y	Valor y (Dual)
331	4.239e+01
332	4.274e+01
333	4.999e+01
334	4.441e+01
335	5.083e+01
336	4.851e+01
337	5.145e+01
338	4.256e+01
339	2.307e+01
340	3.695e+01
341	2.307e+01
342	3.695e+01
343	2.408e+01
344	2.139e+01
345	8.119e+01
346	3.695e+01
347	5.670e+00
348	5.726e+00
349	5.620e+00
350	5.620e+00
351	5.520e+00
352	3.587e+01
353	3.752e+01
354	3.739e+01
355	3.620e+01
356	4.231e+01
357	6.142e+01
358	4.794e+01
359	7.770e+01
360	2.757e+01
361	2.608e+01
362	4.860e+01
363	1.912e+01
364	2.035e+01
365	5.361e+01
366	3.024e+01
367	3.603e+00
368	2.406e+01
369	3.702e+01
370	3.418e+01
371	3.603e+00
372	2.128e+01
373	3.702e+01
374	3.418e+01
375	3.603e+00
376	2.128e+01
377	3.702e+01
378	3.418e+01
379	3.603e+00

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Coordenada y	Valor y (Dual)
380	2.128e+01
381	3.702e+01
382	3.418e+01
383	3.603e+00
384	2.128e+01
385	3.702e+01
386	3.418e+01
387	3.603e+00
388	2.128e+01
389	3.702e+01
390	3.024e+01
391	3.603e+00
392	3.024e+01
393	3.603e+00
394	5.537e+00
395	3.024e+01
396	3.603e+00
397	3.024e+01
398	3.603e+00
399	3.084e+00
401	1.390e-02
402	6.350e-03
403	6.350e-03
404	1.390e-02
406	6.350e-03
407	6.350e-03
408	3.084e+00
410	1.390e-02
412	6.350e-03
413	6.350e-03
414	3.084e+00
416	1.390e-02
418	6.350e-03
419	6.350e-03
420	3.084e+00
422	1.390e-02
423	6.350e-03
424	6.350e-03
425	3.084e+00
427	1.390e-02
429	6.350e-03
430	6.350e-03
431	3.084e+00
433	1.390e-02
434	6.350e-03
435	6.350e-03
436	3.084e+00
438	1.390e-02
439	6.350e-03

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Coordenada y	Valor y (Dual)
440	6.350e-03
441	3.084e+00
443	1.390e-02
444	6.350e-03
445	6.350e-03
446	3.084e+00
448	1.390e-02
450	6.350e-03
451	6.350e-03
452	2.894e+00
453	1.686e-01
454	9.309e-01
455	2.607e+00
456	7.667e-02
457	1.288e+00
458	3.265e+00
459	1.258e-01
461	2.580e+00
462	6.327e-01
463	1.463e-01
464	2.959e+00
465	7.930e-01
467	2.645e+00
469	1.019e+00
470	2.538e+00
471	1.278e-02
472	1.461e+00
473	1.061e+01
474	1.791e+01
475	1.728e+01
476	1.569e+01
478	3.545e+00
481	1.140e-03
482	3.546e+00
486	1.634e+01
487	1.666e+01
488	1.524e+01
489	1.747e+01
490	1.656e+01
491	2.879e+00
492	2.879e+00
493	1.400e+01
494	2.494e+01
495	2.494e+01
496	1.747e+01
497	1.732e+01
498	2.508e+00
499	3.370e+00
500	1.680e+01
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Coordenada y	Valor y (Dual)
502	1.240e+00
503	1.240e+00
505	1.414e+01
506	2.524e+01
507	2.320e+01
508	2.554e+01
509	2.727e+01
511	6.753e-02
514	2.188e+00
515	9.101e+00
517	3.352e+00
518	3.352e+00
519	2.491e+01
520	2.571e+01
521	2.260e+01
522	2.260e+01
523	2.482e+01
524	3.397e+01
525	3.397e+01
526	2.556e+01
527	2.508e+00
528	3.370e+00
530	7.412e+00
531	1.248e+00
532	1.931e+01
533	2.998e+01
534	2.796e+01
535	2.978e+01
536	1.991e+00
537	3.917e+00
540	5.360e+00
541	3.916e+00
542	1.130e-03
543	6.902e+00
544	1.142e+01
545	1.142e+01
546	2.777e+01
547	2.671e+01
548	2.912e+01
549	2.998e+01
550	2.663e+01
551	2.663e+01
552	2.902e+01
553	2.879e+00
554	2.879e+00
555	1.400e+01
556	3.717e+01
557	3.717e+01
558	2.982e+01
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Coordenada y	Valor y (Dual)
559	2.508e+00
560	3.370e+00
561	1.899e+01
563	5.568e+00
564	9.831e+00
565	9.831e+00
567	1.949e+01
568	1.764e+01
569	1.840e+01
570	1.990e+00
571	3.828e+00
572	4.468e-01
573	4.468e-01
574	2.682e+01
575	7.154e+00
576	3.827e+00
577	1.527e+00